

# IT332/CS446 - Assignment 3 Understanding EBS



# Understanding EBS Assignment

- Create new Disk, size is last two digit of your student ID
  - i.e. Student ID = 111111 then volume size is 11G
- New Disk and Mount at /data<ID>
  - i.e Student ID = 111111 then /data111111
- Web Server Run at this Newly Mounted Disk
  - Display Student Name, ID, Machine Name and Disk Path
- Snapshot has been created based on this

# Assignment Review

- Instance summary
- Public IP
- Private IP
- State

**Instance summary for i-040646d074e14f955 (joe111111)** [Info](#)

[Refresh](#) [Connect](#) [Instance state](#) [Actions](#)

Updated less than a minute ago

|                                                                               |                                                                                             |                                                                                                                                            |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Instance ID</b><br><a href="#">i-040646d074e14f955</a>                     | <b>Public IPv4 address</b><br><a href="#">54.145.164.248</a>   <a href="#">open address</a> | <b>Private IPv4 addresses</b><br><a href="#">172.31.84.177</a>                                                                             |
| <b>IPv6 address</b><br>-                                                      | <b>Instance state</b><br><a href="#">Running</a>                                            | <b>Public DNS</b><br><a href="#">ec2-54-145-164-248.compute-1.amazonaws.com</a>   <a href="#">open address</a>                             |
| <b>Hostname type</b><br>IP name: ip-172-31-84-177.ec2.internal                | <b>Private IP DNS name (IPv4 only)</b><br><a href="#">ip-172-31-84-177.ec2.internal</a>     | <b>Elastic IP addresses</b><br>-                                                                                                           |
| <b>Answer private resource DNS name</b><br>-                                  | <b>Instance type</b><br>t3.micro                                                            | <b>AWS Compute Optimizer finding</b><br><a href="#">Opt-in to AWS Compute Optimizer for recommendations.</a><br><a href="#">Learn more</a> |
| <b>Auto-assigned IP address</b><br><a href="#">54.145.164.248</a> [Public IP] | <b>VPC ID</b><br><a href="#">vpc-089e00b8aef4f53f8</a>                                      |                                                                                                                                            |

# Assignment Review

- Volume information
  - Indicated Volume ID
  - Attached resources
  - Disk e.g. /dev/sdf

**vol-01bd71ce1451791b7 (assignment-disk)**

Last updated less than a minute ago [Actions](#) [Delete](#) [Modify](#)

**Details**

|                                                                                                                                                                                                              |                                                                                                                   |                                                                                                                                                          |                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Volume ID</b><br> vol-01bd71ce1451791b7 (assignment-disk)                                                                | <b>Size</b><br> 11 GiB         | <b>Type</b><br>gp3                                                                                                                                       | <b>Status check</b><br> Okay |
| <b>AWS Compute Optimizer finding</b><br> This user is not authorized to call AWS Compute Optimizer.   <a href="#">Retry</a> | <b>Volume state</b><br> In-use | <b>IOPS</b><br>3000                                                                                                                                      | <b>Throughput</b><br>125                                                                                        |
| <b>Fast snapshot restored</b><br>No                                                                                                                                                                          | <b>Availability Zone</b><br>use1-az2 (us-east-1a)                                                                 | <b>Created</b><br> Tue Sep 01 2026 16:29:05 GMT+0700 (Indochina Time) | <b>Multi-Attach enabled</b><br>No                                                                               |
| <b>Attached resources</b><br>i-040646d074e14f955 (joe111111)<br>(EBS card index: 0): /dev/sdf (attached)                                                                                                     | <b>Outposts ARN</b><br>-                                                                                          | <b>Managed</b><br>false                                                                                                                                  | <b>Operator</b><br>-                                                                                            |

# Assignment Review

- Snapshot
  - Source Volume
  - Source Volume Size

**snap-07534a1eedbba2d6 (assignment-snapshot)**  
Last updated less than a minute ago  [Delete](#) [Actions ▼](#)

**Details**

**Snapshot ID**  
 snap-07534a1eedbba2d6 (assignment-snapshot)

**Owner**  
 211125458985

**Description**  
 assignment snapshot

**▼ Source volume**

**Volume ID**  
 vol-01bd71ce1451791b7

**Volume size**  
 11 GiB

**Full snapshot size**  
 253 MiB

**Started**  
 Tue Sep 01 2026 16:49:46 GMT+0700 (Indochina Time)

**Progress**  
 100%

**Product codes**  
-

**Snapshot status**  
 Completed

**Fast snapshot restore**  
-

# Assignment Review

- Command

\$ lsblk

\$ df -h

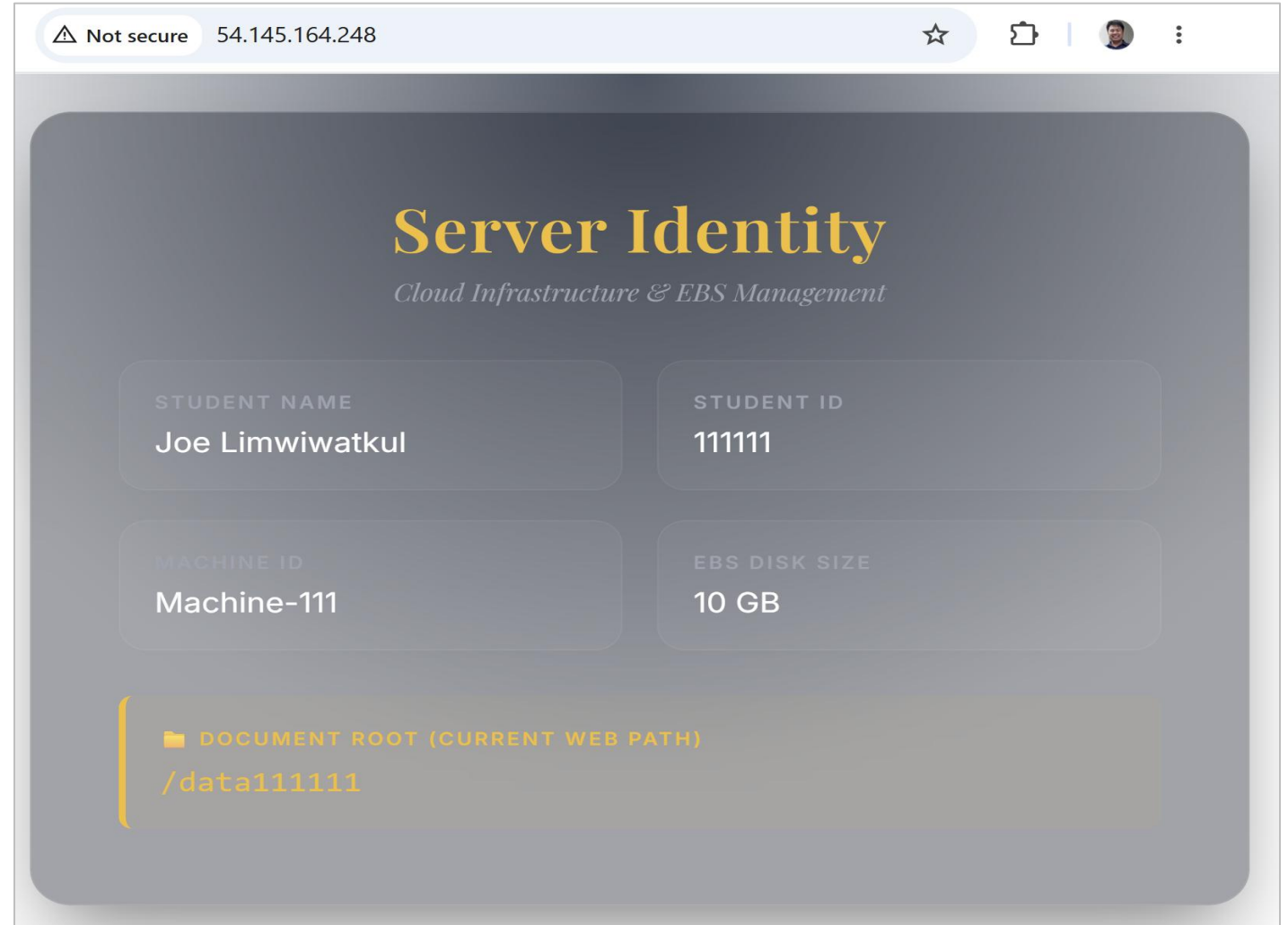
\$ ls -l

/data<ID>

```
ubuntu@ip-172-31-84-177:~$ lsblk
NAME            MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0           7:0      0  28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1           7:1      0  49.3M  1 loop /snap/snapd/26865
loop2           7:2      0   74M  1 loop /snap/core22/2411
nvme0n1         259:0     0    8G   0 disk
├─nvme0n1p1     259:1     0    7G   0 part /
├─nvme0n1p14    259:2     0    4M   0 part
├─nvme0n1p15    259:3     0  106M   0 part /boot/efi
└─nvme0n1p16    259:4     0  913M   0 part /boot
nvme1n1         259:5     0   11G   0 disk /data111111
ubuntu@ip-172-31-84-177:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.8G  2.0G  4.8G  30% /
tmpfs            456M   0    456M   0% /dev/shm
tmpfs            183M  892K  182M   1% /run
tmpfs            5.0M   0    5.0M   0% /run/lock
efivarfs         128K   3.1K  120K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p16  881M   94M  726M  12% /boot
/dev/nvme0n1p15  105M   6.2M   99M   6% /boot/efi
tmpfs            92M   12K   92M   1% /run/user/1000
/dev/nvme1n1     11G   32K   11G   1% /data111111
ubuntu@ip-172-31-84-177:~$ ls -l /data111111
total 24
-rw-r--r-- 1 root root  4361 Sep  1 09:39 index.html
drwx----- 2 root root 16384 Sep  1 09:30 lost+found
ubuntu@ip-172-31-84-177:~$
```

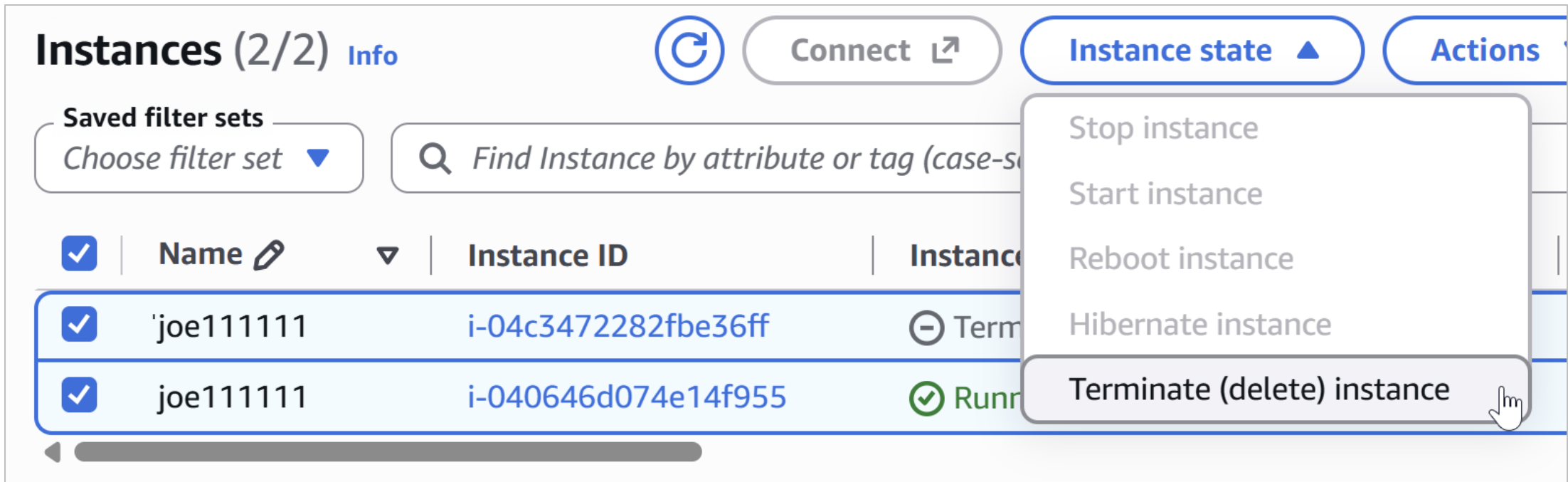
# Assignment Review

- Web Server
- IP = Public IP Address
- Present
  - Name
  - ID
  - Machine ID
  - Document Root



# Cleanup the Instance

- Select any instance
- Instance state → Terminate (delete) instance

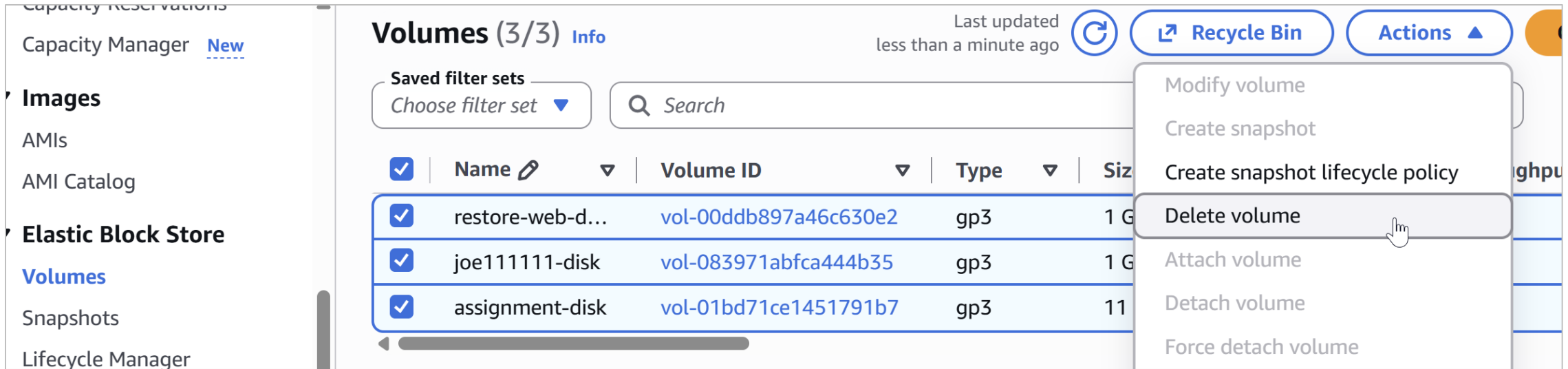


The screenshot displays the AWS Management Console's 'Instances' page. At the top, it shows 'Instances (2/2)' with an 'Info' link. Below this are filter controls, including 'Saved filter sets' and a search bar. A table lists two instances, both named 'joe111111'. The first instance has ID 'i-04c3472282fbe36ff' and a 'Terminated' state. The second instance has ID 'i-040646d074e14f955' and a 'Running' state. An 'Actions' dropdown menu is open for the 'Running' instance, showing options: 'Stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'. A mouse cursor is pointing at the 'Terminate (delete) instance' option.

|                                     | Name      | Instance ID         | Instance State |
|-------------------------------------|-----------|---------------------|----------------|
| <input checked="" type="checkbox"/> | joe111111 | i-04c3472282fbe36ff | Terminated     |
| <input checked="" type="checkbox"/> | joe111111 | i-040646d074e14f955 | Running        |

# Cleanup Volume

- Elastic Block Store → Volumes
- Select All → Actions → Delete volume



The screenshot shows the AWS Management Console interface for the Elastic Block Store (EBS) Volumes page. The left sidebar contains navigation links for Capacity Reservations, Capacity Manager, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, and Lifecycle Manager. The main content area is titled 'Volumes (3/3)' and includes a 'Last updated' status of 'less than a minute ago'. Below the title is a search bar and a table of volumes. The table has columns for Name, Volume ID, Type, and Size. Three volumes are listed: 'restore-web-d...', 'joe111111-disk', and 'assignment-disk'. An 'Actions' button is visible in the top right, and a dropdown menu is open, showing options like 'Modify volume', 'Create snapshot', 'Delete volume' (which is highlighted), 'Attach volume', 'Detach volume', and 'Force detach volume'.

|                                     | Name             | Volume ID             | Type | Size |
|-------------------------------------|------------------|-----------------------|------|------|
| <input checked="" type="checkbox"/> | restore-web-d... | vol-00ddb897a46c630e2 | gp3  | 1 G  |
| <input checked="" type="checkbox"/> | joe111111-disk   | vol-083971abfca444b35 | gp3  | 1 G  |
| <input checked="" type="checkbox"/> | assignment-disk  | vol-01bd71ce1451791b7 | gp3  | 11   |

# Cleanup Snapshot

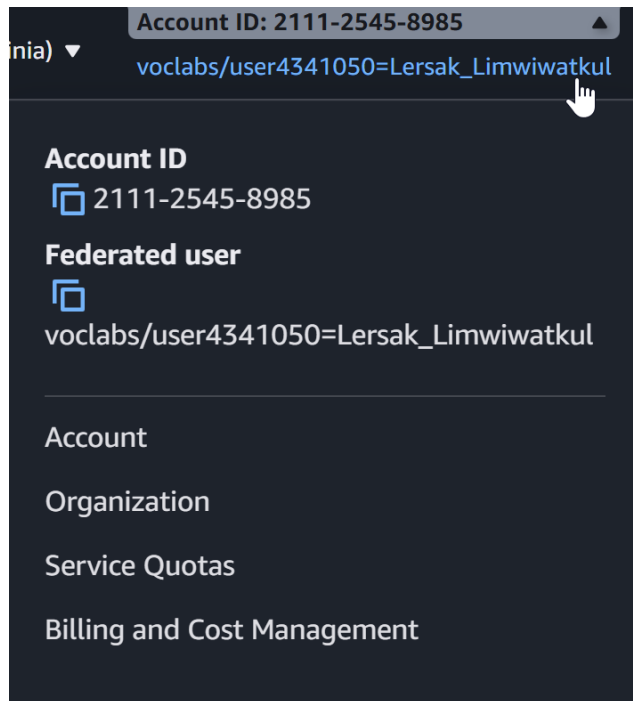
- Elastic Block Store → Snapshots
- Select All → Actions → Delete snapshot

The screenshot shows the AWS Management Console interface for the 'Snapshots' section. On the left, a navigation pane lists 'AMI Catalog', 'Elastic Block Store' (with sub-items 'Volumes' and 'Snapshots'), and 'Network & Security' (with sub-item 'Security Groups'). The main content area is titled 'Snapshots (2/2)' and includes a 'Last updated' status of 'less than a minute ago'. Below this, there's a 'Snapshot scope' dropdown set to 'Owned by me' and a search bar. A table lists two snapshots, both of which are selected with checkboxes. An 'Actions' button is visible, and its dropdown menu is open, showing options like 'Create volume from snapshot', 'Create image from snapshot', 'Copy snapshot', 'Launch copy duration calculator', 'Delete snapshot' (which is highlighted by a mouse cursor), and 'Manage tags'.

|                                     | Name             | Snapshot ID            | Full snapshot size |
|-------------------------------------|------------------|------------------------|--------------------|
| <input checked="" type="checkbox"/> | assignment-sn... | snap-07534a1eedbba2d6  | 253 MiB            |
| <input checked="" type="checkbox"/> | backup-web-d...  | snap-0650a84c09ad1a860 | 52 MiB             |

# Billings and Cost Management

- Billing and Cost Management Home



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### Cost summary [Info](#)

|                                                                                                           |                                                                             |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Month-to-date cost<br><b><u>\$0.11</u></b><br>- compared to last month for same period                    | Last month's cost for same time period<br><b><u>\$0.00</u></b><br>Aug 1 – 6 |
| Total forecasted cost for current month<br><b><u>\$0.17</u></b><br>- compared to last month's total costs | Last month's total cost<br><b><u>\$0.00</u></b>                             |

[View bill](#)

# Sign out and End Lab

- Sign out from AWS Console

- Click End Lab

